Lesson 9: Probability Spaces and Conditioning

AI/ML Mathematics

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1 Motivation and Overview

Consider a reinforcement learning agent navigating a stochastic gridworld. At each time step the agent selects an action, but the resulting state transition is random: a "move north" command might succeed with probability 0.8 and slip sideways with probability 0.1 in each perpendicular direction. To compute expected returns, compare policies, or update beliefs after observing a trajectory, the agent needs a rigorous language for quantifying uncertainty. What does it mean for an event to have probability 0.8? How should the agent revise its beliefs about a hidden parameter after collecting data? If two noise sources affect the system, when can their contributions be analyzed separately?

The tools developed in this lesson will let us formalize these questions. We construct probability spaces via the Kolmogorov axioms, derive the fundamental rules for computing with probabilities (complement, monotonicity, inclusion-exclusion, continuity), and develop conditional probability, Bayes' theorem, and the concept of independence. These results form the prerequisite language for every probabilistic statement in reinforcement learning and machine learning.

Why Probability Spaces Matter for RL/ML

- **Markov decision processes**: the transition kernel $p(s' \mid s, a)$ is a conditional probability; the Markov property is a conditional independence statement.
- **Bayesian inference**: posterior updates in Thompson sampling and Bayesian neural networks apply Bayes' theorem at every step.
- **Monte Carlo methods**: REINFORCE and Monte Carlo tree search estimate expectations by sampling from probability spaces.
- **PAC learning bounds**: generalization guarantees in supervised learning rely on the union bound, inclusion-exclusion, and independence of training samples.

Prerequisites. Basic set theory: unions, intersections, complements, De Morgan's laws (Lesson 1). Familiarity with sequences and limits of real numbers (Lessons 2–3). No linear algebra is required for this lesson.

Learning objectives. After completing this lesson, the student will be able to:

- *Define* sample spaces, events, and probability measures via the Kolmogorov axioms.
- *Derive* basic properties of probability (complement rule, monotonicity, inclusion-exclusion) from the axioms.
- *Prove* the continuity of probability for monotone sequences of events.
- *Compute* conditional probabilities and apply the multiplication rule, law of total probability, and Bayes' theorem.
- *Distinguish* between pairwise and mutual independence and *construct* examples separating the two notions.
- *Apply* Bayes' theorem to belief-update problems arising in machine learning.

2 References

- Ross [2019, Chapters 1–2] – Combinatorial analysis, axioms of probability, conditional probability and independence.

- Blitzstein and Hwang [2019, Chapters 1–2] – Probability and counting, conditional probability with extensive examples.
- Feller [1968, Chapter 1] – Classical introduction to sample spaces and probability axioms.
- Grimmett and Stirzaker [2020, Chapter 1] – Events and their probabilities, with a path toward measure-theoretic foundations.
- Bertsekas and Tsitsiklis [2008, Chapters 1–2] – Sample space, probability axioms, conditioning, and independence with engineering applications.

3 Probability Spaces

We begin by defining the mathematical objects that model random experiments.

Definition 3.1 (Sample Space, Outcome, Event)**.** A *sample space* Ω is a non-empty set. Each element $\omega \in \Omega$ is called an *outcome*. An *event* is a subset $A \subseteq \Omega$.

Remark 3.2. In this lesson we take the collection of all events to be the power set 2^Ω (the set of all subsets of Ω). For uncountable sample spaces, one must restrict to a σ-algebra $\mathcal{F} \subseteq 2^\Omega$ to avoid measure-theoretic paradoxes; this refinement is developed in Phase 01 (Lesson 1, σ-algebras).

Definition 3.3 (Set Operations on Events)**.** Let $A, B \subseteq \Omega$.

1. *Union:* $A \cup B = \{\omega \in \Omega : \omega \in A \text{ or } \omega \in B\}$ ("A or B occurs").
2. *Intersection:* $A \cap B = \{\omega \in \Omega : \omega \in A \text{ and } \omega \in B\}$ ("both A and B occur").
3. *Complement:* $A^c = \Omega \setminus A = \{\omega \in \Omega : \omega \notin A\}$ ("A does not occur").
4. *Disjoint:* A and B are *disjoint* (or *mutually exclusive*) if $A \cap B = \emptyset$.

Example 3.4 (Die Roll)**.** Roll a fair six-sided die. The sample space is $\Omega = \{1, 2, 3, 4, 5, 6\}$. Let $A = \{2, 4, 6\}$ ("even") and $B = \{1, 2, 3\}$ ("at most 3"). Then $A \cup B = \{1, 2, 3, 4, 6\}$, $A \cap B = \{2\}$, and $A^c = \{1, 3, 5\}$.

3.1 The Kolmogorov Axioms

Definition 3.5 (Probability Measure – Kolmogorov Axioms)**.** A *probability measure* (or simply a *probability*) on a sample space Ω is a function $\mathbb{P}\colon 2^\Omega \to \mathbb{R}$ satisfying:
(K1) **Non-negativity:** $\mathbb{P}(A) \geq 0$ for every event A.
(K2) **Normalization:** $\mathbb{P}(\Omega) = 1$.
(K3) **Countable additivity:** If $A_1, A_2, \ldots$ are pairwise disjoint events, then

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n).$$

The pair $(\Omega, \mathbb{P})$ is called a *probability space*.

Remark 3.6. Countable additivity (K3) implies *finite additivity*: if $A_1, \ldots, A_n$ are pairwise disjoint, set $A_k = \emptyset$ for $k > n$. Once we prove $\mathbb{P}(\emptyset) = 0$ below, the tail terms vanish and $\mathbb{P}(\bigcup_{i=1}^n A_i) = \sum_{i=1}^n \mathbb{P}(A_i)$.

Example 3.7 (Uniform Probability on a Finite Set)**.** Let Ω be a finite set with $|\Omega| = N$. Define $\mathbb{P}(A) = |A|/N$ for every $A \subseteq \Omega$. Then (K1) holds because $|A| \geq 0$; (K2) holds because $|\Omega|/N = 1$; (K3) holds because for disjoint sets $|A \cup B| = |A| + |B|$ (and only finitely many events are non-empty, so the infinite sum reduces to a finite one).

3.2 Basic Properties

We now derive the fundamental consequences of the Kolmogorov axioms.

Proposition 3.8 (Complement Rule). *For any event A, $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.*

Proof. Since A and A^c are disjoint and $A \cup A^c = \Omega$, countable additivity (K3) (applied to two events, with the remaining sets empty) gives $\mathbb{P}(\Omega) = \mathbb{P}(A) + \mathbb{P}(A^c)$. By normalization (K2), $\mathbb{P}(\Omega) = 1$, so $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$. □

Corollary 3.9 ($\mathbb{P}(\emptyset) = 0$). $\mathbb{P}(\emptyset) = 0$.

Proof. Since $\emptyset = \Omega^c$, the complement rule gives $\mathbb{P}(\emptyset) = 1 - \mathbb{P}(\Omega) = 1 - 1 = 0$. □

Proposition 3.10 (Monotonicity). *If $A \subseteq B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$.*

Proof. Write $B = A \cup (B \setminus A)$, where A and $B \setminus A$ are disjoint. By finite additivity, $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A)$. Since $\mathbb{P}(B \setminus A) \geq 0$ by (K1), we conclude $\mathbb{P}(B) \geq \mathbb{P}(A)$. □

Example 3.11 (Applying Monotonicity). Roll a fair die. Let $A = \{6\}$ and $B = \{4, 5, 6\}$. Since $A \subseteq B$, monotonicity guarantees $\mathbb{P}(A) \leq \mathbb{P}(B)$. Indeed, $1/6 \leq 3/6 = 1/2$.

Inclusion-Exclusion Principle

The inclusion-exclusion formula generalizes finite additivity to overlapping events. For two events it corrects for double-counting; for n events it alternates between over- and under-counting.

Theorem 3.12 (Inclusion-Exclusion for Two Events). *For any events A and B,*

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

Proof. Write $A \cup B = A \cup (B \setminus A)$, where A and $B \setminus A$ are disjoint. By finite additivity,

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A). \tag{1}$$

Similarly, $B = (A \cap B) \cup (B \setminus A)$ with $A \cap B$ and $B \setminus A$ disjoint, so

$$\mathbb{P}(B) = \mathbb{P}(A \cap B) + \mathbb{P}(B \setminus A), \tag{2}$$

which gives $\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A \cap B)$. Substituting into (1):

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$ □

Example 3.13 (Inclusion-Exclusion Applied). Draw one card from a standard 52-card deck. Let A = "the card is a heart" and B = "the card is a face card (J, Q, K)." Then $\mathbb{P}(A) = 13/52$, $\mathbb{P}(B) = 12/52$, and $\mathbb{P}(A \cap B) = 3/52$ (the three heart face cards). By inclusion-exclusion, $\mathbb{P}(A \cup B) = 13/52 + 12/52 - 3/52 = 22/52 = 11/26$.

Theorem 3.14 (Inclusion-Exclusion for n Events). *For events $A_1, \ldots, A_n$,*

$$\mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{k=1}^{n} (-1)^{k+1} \sum_{1 \leq i_1 < \cdots < i_k \leq n} \mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k}).$$

Proof. We proceed by induction on n.

Base case ($n = 1$): $\mathbb{P}(A_1) = \mathbb{P}(A_1)$, which is trivially the formula with a single term.

Base case ($n = 2$): This is Theorem 3.12.

Inductive step. Assume the formula holds for $n - 1$ events. Write

$$\bigcup_{i=1}^{n} A_i = \left(\bigcup_{i=1}^{n-1} A_i\right) \cup A_n.$$

By Theorem 3.12,

$$\mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) = \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) + \mathbb{P}(A_n) - \mathbb{P}\left(\left(\bigcup_{i=1}^{n-1} A_i\right) \cap A_n\right). \tag{3}$$

By distributivity, $\left(\bigcup_{i=1}^{n-1} A_i\right) \cap A_n = \bigcup_{i=1}^{n-1}(A_i \cap A_n)$. Apply the inductive hypothesis to the first term in (3) (a union of $n - 1$ events $A_1, \ldots, A_{n-1}$) and to the third term (a union of $n - 1$ events $A_1 \cap A_n, \ldots, A_{n-1} \cap A_n$).

The first term contributes the alternating sum over all subsets of $\{1, \ldots, n-1\}$. Adding $\mathbb{P}(A_n)$ accounts for the singleton $\{n\}$. Subtracting the inductive hypothesis applied to $\bigcup_{i=1}^{n-1}(A_i \cap A_n)$ contributes exactly the terms involving A_n in subsets of size ≥ 2, each with sign $(-1)^{k+1}$ (the extra intersection with A_n increases the subset size by one and flips the sign). Collecting all terms yields the formula for n events. □

Union Bound as a Corollary

A common shortcut is the *union bound* (Boole's inequality): $\mathbb{P}(\bigcup_{i=1}^{n} A_i) \leq \sum_{i=1}^{n} \mathbb{P}(A_i)$. This follows from inclusion-exclusion by dropping the alternating correction terms (or directly from monotonicity and finite additivity). In machine learning, the union bound is the workhorse behind PAC learning arguments and uniform convergence bounds. Be aware that it can be very loose when events overlap significantly.

3.3 Continuity of Probability

The following theorem connects the axioms to limits of sequences of events and is essential for analyzing asymptotic properties.

Theorem 3.15 (Continuity of Probability). *(a)* ***Continuity from below.*** *If $A_1 \subseteq A_2 \subseteq A_3 \subseteq \cdots$ (increasing), then*

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} \mathbb{P}(A_n).$$

(b) ***Continuity from above.*** *If $A_1 \supseteq A_2 \supseteq A_3 \supseteq \cdots$ (decreasing), then*

$$\mathbb{P}\left(\bigcap_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} \mathbb{P}(A_n).$$

Proof. **(a)** Define $B_1 = A_1$ and $B_n = A_n \setminus A_{n-1}$ for $n \geq 2$. The events $B_1, B_2, \ldots$ are pairwise disjoint: if $i < j$, then $B_i \subseteq A_i \subseteq A_{j-1}$, but $B_j = A_j \setminus A_{j-1}$ is disjoint from A_{j-1}, so $B_i \cap B_j = \emptyset$. Moreover, $\bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} A_n$. By countable additivity (K3),

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(B_n).$$

The partial sums satisfy $\sum_{k=1}^{n} \mathbb{P}(B_k) = \mathbb{P}(\bigcup_{k=1}^{n} B_k) = \mathbb{P}(A_n)$, where the first equality uses finite additivity (the B_k are disjoint) and the second uses $\bigcup_{k=1}^{n} B_k = A_n$ (proved by induction: $B_1 = A_1$, and adding $B_n = A_n \setminus A_{n-1}$ to A_{n-1} gives A_n). Therefore

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} \sum_{k=1}^{n} \mathbb{P}(B_k) = \lim_{n\to\infty} \mathbb{P}(A_n).$$

(b) If $A_1 \supseteq A_2 \supseteq \cdots$, then the complements $A_1^c \subseteq A_2^c \subseteq \cdots$ form an increasing sequence. By part (a),

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n^c\right) = \lim_{n\to\infty} \mathbb{P}(A_n^c) = \lim_{n\to\infty} (1 - \mathbb{P}(A_n)).$$

By De Morgan's law, $\bigcup_{n=1}^{\infty} A_n^c = (\bigcap_{n=1}^{\infty} A_n)^c$, so the left side equals $1 - \mathbb{P}(\bigcap_{n=1}^{\infty} A_n)$. Equating and simplifying: $\mathbb{P}(\bigcap_{n=1}^{\infty} A_n) = \lim_{n\to\infty} \mathbb{P}(A_n)$. □

Example 3.16 (Continuity from Below)**.** Let $\Omega = (0, 1]$ with the uniform distribution (Lebesgue measure). Define $A_n = (0, 1 - 1/n]$ for $n \geq 2$. Then $A_2 \subseteq A_3 \subseteq \cdots$ and $\bigcup_{n=2}^{\infty} A_n = (0, 1)$. By continuity from below, $\mathbb{P}((0, 1)) = \lim_{n\to\infty} \mathbb{P}(A_n) = \lim_{n\to\infty}(1 - 1/n) = 1$. This confirms that $\mathbb{P}(\{1\}) = 0$ for a continuous uniform distribution.

Continuity and Convergence in RL

Continuity from above is essential for "almost sure" convergence statements in RL. The event "the value estimate V_t converges to the true value V^π" is expressed as $\bigcap_{k=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{t=N}^{\infty} \{|V_t - V^\pi| < 1/k\}$. Computing the probability of such nested limits requires the continuity of $\mathbb{P}$ along monotone sequences.

We have established the axiomatic framework and its basic consequences. The next section introduces conditional probability, which allows us to update probabilities given partial information.

4 Conditional Probability

In many situations we have partial information about the outcome of an experiment, and we wish to update our probability assignments accordingly. Conditional probability formalizes this update.

Definition 4.1 (Conditional Probability)**.** Let $(\Omega, \mathbb{P})$ be a probability space and let B be an event with $\mathbb{P}(B) > 0$. The *conditional probability of A given B* is

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Remark 4.2. For a fixed event B with $\mathbb{P}(B) > 0$, the function $A \mapsto \mathbb{P}(A \mid B)$ is itself a probability measure on Ω. Verification: (K1) $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B) \geq 0$; (K2) $\mathbb{P}(\Omega \mid B) = \mathbb{P}(B)/\mathbb{P}(B) = 1$; (K3) if $A_1, A_2, \ldots$ are pairwise disjoint, then $\mathbb{P}(\bigcup_n A_n \mid B) = \mathbb{P}((\bigcup_n A_n) \cap B)/\mathbb{P}(B) = \mathbb{P}(\bigcup_n (A_n \cap B))/\mathbb{P}(B) = \sum_n \mathbb{P}(A_n \cap B)/\mathbb{P}(B) = \sum_n \mathbb{P}(A_n \mid B)$.

Example 4.3 (Conditional Probability with Dice)**.** Roll two fair dice. Let A = "the sum is 8" and B = "the first die shows 3." The sample space has $|\Omega| = 36$ equally likely outcomes. We have $A \cap B = \{(3, 5)\}$, so $\mathbb{P}(A \cap B) = 1/36$, and $\mathbb{P}(B) = 6/36 = 1/6$. Thus $\mathbb{P}(A \mid B) = (1/36)/(1/6) = 1/6$.

4.1 Multiplication Rule

The multiplication rule extends the definition of conditional probability to chains of events.

Theorem 4.4 (Multiplication Rule)**.** *For events $A_1, \ldots, A_n$ with $\mathbb{P}(A_1 \cap \cdots \cap A_{n-1}) > 0$,*

$$\mathbb{P}(A_1 \cap \cdots \cap A_n) = \mathbb{P}(A_1)\,\mathbb{P}(A_2 \mid A_1)\,\mathbb{P}(A_3 \mid A_1 \cap A_2) \cdots \mathbb{P}(A_n \mid A_1 \cap \cdots \cap A_{n-1}).$$

Proof. We proceed by induction on n.

Base case ($n = 2$): By definition of conditional probability, $\mathbb{P}(A_1 \cap A_2) = \mathbb{P}(A_1)\,\mathbb{P}(A_2 \mid A_1)$, since $\mathbb{P}(A_2 \mid A_1) = \mathbb{P}(A_1 \cap A_2)/\mathbb{P}(A_1)$ and $\mathbb{P}(A_1) \geq \mathbb{P}(A_1 \cap A_2) > 0$ by monotonicity (as $A_1 \cap A_2 \subseteq A_1 \cap \cdots \cap A_{n-1}$ and $\mathbb{P}(A_1 \cap \cdots \cap A_{n-1}) > 0$ implies $\mathbb{P}(A_1) > 0$).

Inductive step. Assume the formula holds for $n-1$ events. Since $\mathbb{P}(A_1 \cap \cdots \cap A_{n-1}) > 0$, we can write

$$\begin{aligned}
\mathbb{P}(A_1 \cap \cdots \cap A_n) &= \mathbb{P}(A_1 \cap \cdots \cap A_{n-1}) \cdot \mathbb{P}(A_n \mid A_1 \cap \cdots \cap A_{n-1}) \\
&= \Big[\mathbb{P}(A_1)\,\mathbb{P}(A_2 \mid A_1) \cdots \mathbb{P}(A_{n-1} \mid A_1 \cap \cdots \cap A_{n-2})\Big] \\
&\quad \cdot \mathbb{P}(A_n \mid A_1 \cap \cdots \cap A_{n-1}),
\end{aligned}$$

where the first line uses the definition of conditional probability and the second applies the inductive hypothesis to the $n-1$ events $A_1, \ldots, A_{n-1}$. □

Example 4.5 (Drawing Without Replacement)**.** An urn contains 5 red and 3 blue balls. We draw 3 balls without replacement. Let A_i = "the i-th ball drawn is red." By the multiplication rule,

$$\begin{aligned}
\mathbb{P}(A_1 \cap A_2 \cap A_3) &= \mathbb{P}(A_1)\,\mathbb{P}(A_2 \mid A_1)\,\mathbb{P}(A_3 \mid A_1 \cap A_2) \\
&= \frac{5}{8} \cdot \frac{4}{7} \cdot \frac{3}{6} = \frac{60}{336} = \frac{5}{28}.
\end{aligned}$$

4.2 Law of Total Probability

Definition 4.6 (Partition)**.** Events $B_1, B_2, \ldots, B_n$ form a *partition* of Ω if they are pairwise disjoint and $\bigcup_{i=1}^n B_i = \Omega$.

Theorem 4.7 (Law of Total Probability)**.** *Let $B_1, B_2, \ldots, B_n$ be a partition of Ω with $\mathbb{P}(B_i) > 0$ for all i. Then for any event A,*

$$\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(A \mid B_i)\,\mathbb{P}(B_i).$$

Proof. Since the B_i partition Ω,

$$A = A \cap \Omega = A \cap \left(\bigcup_{i=1}^n B_i\right) = \bigcup_{i=1}^n (A \cap B_i).$$

The events $A \cap B_1, \ldots, A \cap B_n$ are pairwise disjoint (since the B_i are). By finite additivity,

$$\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(A \cap B_i) = \sum_{i=1}^n \mathbb{P}(A \mid B_i)\,\mathbb{P}(B_i),$$

where the last equality uses $\mathbb{P}(A \cap B_i) = \mathbb{P}(A \mid B_i)\,\mathbb{P}(B_i)$ from the definition of conditional probability. □

Example 4.8 (Medical Test – Total Probability)**.** A disease affects 1% of the population. A diagnostic test has a sensitivity (true positive rate) of 95% and a specificity (true negative rate) of 90%. Let D = "has disease" and T = "tests positive." The partition is $\{D, D^c\}$ with $\mathbb{P}(D) = 0.01$ and $\mathbb{P}(D^c) = 0.99$. By the law of total probability,

$$\begin{aligned}\mathbb{P}(T) &= \mathbb{P}(T \mid D)\,\mathbb{P}(D) + \mathbb{P}(T \mid D^c)\,\mathbb{P}(D^c)\\ &= 0.95 \times 0.01 + 0.10 \times 0.99\\ &= 0.0095 + 0.099 = 0.1085.\end{aligned}$$

Despite the test's high sensitivity, most positive results are false positives because the disease is rare.

4.3 Bayes' Theorem

Bayes' Theorem – The Belief Update Rule

Bayes' theorem inverts the direction of conditioning: given a partition into "hypotheses" $B_1, \ldots, B_n$ with prior probabilities $\mathbb{P}(B_j)$, observing event A updates each hypothesis to the posterior $\mathbb{P}(B_j \mid A)$.

Theorem 4.9 (Bayes' Theorem)**.** *Let $B_1, \ldots, B_n$ be a partition of Ω with $\mathbb{P}(B_i) > 0$ for all i, and let A be an event with $\mathbb{P}(A) > 0$. Then for each $j = 1, \ldots, n$,*

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\,\mathbb{P}(B_j)}{\sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\,\mathbb{P}(B_i)}.$$

Proof. By the definition of conditional probability,

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \cap B_j)}{\mathbb{P}(A)}.$$

The numerator equals $\mathbb{P}(A \mid B_j)\,\mathbb{P}(B_j)$ by the definition of $\mathbb{P}(A \mid B_j)$. The denominator equals $\sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\,\mathbb{P}(B_i)$ by the law of total probability (Theorem 4.7). Substituting gives the result. □

Example 4.10 (Medical Test – Bayes' Theorem)**.** Continuing Example 4.8, suppose a patient tests positive. What is the probability they actually have the disease? By Bayes' theorem,

$$\mathbb{P}(D \mid T) = \frac{\mathbb{P}(T \mid D)\,\mathbb{P}(D)}{\mathbb{P}(T)} = \frac{0.95 \times 0.01}{0.1085} \approx 0.0876.$$

Despite a positive test result, the probability of disease is only about 8.8%. This illustrates the *base rate fallacy*: ignoring the low prior $\mathbb{P}(D)$ leads to grossly overestimating $\mathbb{P}(D \mid T)$.

Example 4.11 (Urn Problem – Bayesian Update)**.** There are two urns. Urn 1 has 7 red and 3 blue balls; Urn 2 has 4 red and 6 blue balls. One urn is chosen uniformly at random, and a ball is drawn. Let U_i = "Urn i was chosen" and R = "a red ball is drawn." The prior is $\mathbb{P}(U_1) = \mathbb{P}(U_2) = 1/2$. We have $\mathbb{P}(R \mid U_1) = 7/10$ and $\mathbb{P}(R \mid U_2) = 4/10$.

Given that a red ball was drawn, Bayes' theorem gives

$$\mathbb{P}(U_1 \mid R) = \frac{(7/10)(1/2)}{(7/10)(1/2) + (4/10)(1/2)} = \frac{7/20}{11/20} = \frac{7}{11} \approx 0.636.$$

The observation of red shifts our belief toward Urn 1.

Bayes' Theorem in RL/ML

- In **Bayesian reinforcement learning**, the agent maintains a prior over transition models and updates it via Bayes' theorem after each observed transition.
- In **POMDPs**, the belief state $b'(s') \propto \mathbb{P}(o \mid s') \sum_s \mathbb{P}(s' \mid s, a)\, b(s)$ is a direct application of the law of total probability and Bayes' theorem.
- In **naive Bayes classifiers**, the class posterior is computed via Bayes' theorem with a conditional independence assumption on features.

We now turn to the concept of independence, which characterizes when knowledge of one event provides no information about another.

5 Independence

Definition 5.1 (Independence of Two Events)**.** Events A and B are *independent* if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B).$$

Remark 5.2. When $\mathbb{P}(B) > 0$, independence is equivalent to $\mathbb{P}(A \mid B) = \mathbb{P}(A)$: learning that B occurred does not change the probability of A. This follows directly from the definition: $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B) = \mathbb{P}(A)\mathbb{P}(B)/\mathbb{P}(B) = \mathbb{P}(A)$.

Example 5.3 (Independent Events)**.** Toss a fair coin twice. Let A = "first toss is heads" and B = "second toss is heads." The sample space is $\Omega = \{HH, HT, TH, TT\}$ with each outcome having probability 1/4. Then $\mathbb{P}(A) = 1/2$, $\mathbb{P}(B) = 1/2$, and $\mathbb{P}(A \cap B) = \mathbb{P}(\{HH\}) = 1/4 = (1/2)(1/2) = \mathbb{P}(A)\mathbb{P}(B)$. So A and B are independent.

Independent $\neq$ Disjoint

A common mistake is to confuse independence with disjointness. If A and B are disjoint and both have positive probability, then $\mathbb{P}(A \cap B) = 0 \neq \mathbb{P}(A)\mathbb{P}(B) > 0$, so they are *not* independent. Disjoint events are maximally dependent: knowing one occurred tells you the other did not.

Proposition 5.4 (Independence of Complements)**.** *If A and B are independent, then so are A and B^c, A^c and B, and A^c and B^c.*

Proof. We prove that A and B^c are independent; the other cases follow by symmetry and iteration. Since $A = (A \cap B) \cup (A \cap B^c)$ with the two sets disjoint,

$$\mathbb{P}(A \cap B^c) = \mathbb{P}(A) - \mathbb{P}(A \cap B) = \mathbb{P}(A) - \mathbb{P}(A)\mathbb{P}(B) = \mathbb{P}(A)(1 - \mathbb{P}(B)) = \mathbb{P}(A)\mathbb{P}(B^c).$$

For A^c and B^c: since A and B^c are independent (just proved), applying the same argument with A replaced by A^c gives $\mathbb{P}(A^c \cap B^c) = \mathbb{P}(A^c)\mathbb{P}(B^c)$. □

5.1 Mutual Independence

Definition 5.5 (Mutual Independence)**.** Events $A_1, \ldots, A_n$ are *mutually independent* if for every subset $\{i_1, \ldots, i_k\} \subseteq \{1, \ldots, n\}$ with $k \geq 2$,

$$\mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k}) = \mathbb{P}(A_{i_1}) \cdots \mathbb{P}(A_{i_k}).$$

This requires the product formula to hold for *all* $\binom{n}{2} + \binom{n}{3} + \cdots + \binom{n}{n} = 2^n - n - 1$ subsets of size ≥ 2, not merely for the $\binom{n}{2}$ pairs.

Remark 5.6. Events $A_1, \ldots, A_n$ are *pairwise independent* if every pair is independent: $\mathbb{P}(A_i \cap A_j) = \mathbb{P}(A_i)\mathbb{P}(A_j)$ for all $i \neq j$. Mutual independence implies pairwise independence (take $k = 2$ in the definition), but the converse is false.

Pairwise $\neq$ Mutual Independence

Pairwise independence does not imply mutual independence. The product rule can hold for every pair yet fail for a triple (or larger subset). This distinction matters in practice: many concentration inequalities (e.g., the variance of a sum) require only pairwise independence, while others (e.g., the strong law of large numbers) require mutual independence.

Example 5.7 (Pairwise but Not Mutually Independent)**.** Toss two fair coins. The sample space is $\Omega = \{HH, HT, TH, TT\}$, each outcome with probability $1/4$. Define:

- A = "the first coin is heads" $= \{HH, HT\}$,
- B = "the second coin is heads" $= \{HH, TH\}$,
- C = "exactly one head" $= \{HT, TH\}$.

Each event has probability $1/2$. We check all pairs:

- $A \cap B = \{HH\}$, so $\mathbb{P}(A \cap B) = 1/4 = (1/2)(1/2) = \mathbb{P}(A)\mathbb{P}(B)$. ✓
- $A \cap C = \{HT\}$, so $\mathbb{P}(A \cap C) = 1/4 = (1/2)(1/2) = \mathbb{P}(A)\mathbb{P}(C)$. ✓
- $B \cap C = \{TH\}$, so $\mathbb{P}(B \cap C) = 1/4 = (1/2)(1/2) = \mathbb{P}(B)\mathbb{P}(C)$. ✓

All three pairs are independent. However, $A \cap B \cap C = \emptyset$ (both heads is not exactly one head), so

$$\mathbb{P}(A \cap B \cap C) = 0 \neq \frac{1}{8} = \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C).$$

Therefore A, B, C are pairwise independent but *not* mutually independent. Knowing A and B both occurred completely determines C (it cannot occur), even though each pair individually reveals nothing.

Example 5.8 (Mutual Independence in Repeated Trials)**.** Toss a fair coin three times independently. The sample space has $|\Omega| = 8$ equally likely outcomes. Let A_i = "the i-th toss is heads" for $i = 1, 2, 3$. Then $\mathbb{P}(A_i) = 1/2$ for each i. For pairs: $\mathbb{P}(A_i \cap A_j) = 2/8 = 1/4 = (1/2)(1/2)$ for each pair $i \neq j$ (two of the eight outcomes have heads on both specified positions). For the triple: $\mathbb{P}(A_1 \cap A_2 \cap A_3) = 1/8 = (1/2)^3$. All $2^3 - 3 - 1 = 4$ conditions are satisfied, confirming mutual independence.

Independence and the Markov Property

The Markov property is fundamentally a conditional independence statement: the future evolution of the process is conditionally independent of the past given the present state. Formally, in a Markov decision process, $\mathbb{P}(S_{t+1} \mid S_0, A_0, \ldots, S_t, A_t) = \mathbb{P}(S_{t+1} \mid S_t, A_t)$. Independence of noise terms across time steps justifies treating transition randomness as i.i.d., which is central to the convergence analysis of TD-learning and Q-learning.

6 Exercises

Exercise 6.1 (Bonferroni Inequalities)**.** $(*)$
Using inclusion-exclusion (Theorem 3.14), prove the *first Bonferroni inequality*: for any events $A_1, \ldots, A_n$,

$$\mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{n} \mathbb{P}(A_i).$$

Then prove the *second Bonferroni inequality*:

$$\mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) \geq \sum_{i=1}^{n} \mathbb{P}(A_i) - \sum_{1 \leq i < j \leq n} \mathbb{P}(A_i \cap A_j).$$

Hint: For the first, use induction or monotonicity. For the second, use inclusion-exclusion truncated at $k = 2$ and show the remaining terms contribute non-negatively.

Exercise 6.2 (Continuity from Above). $(*)$
Let $A_1 \supseteq A_2 \supseteq \cdots$ with $\mathbb{P}(A_n) = 1/(n+1)$ for all $n \geq 1$. Compute $\mathbb{P}(\bigcap_{n=1}^{\infty} A_n)$ using the continuity of probability.

Exercise 6.3 (Sequential Bayes Updates). $(**)$
A factory has two machines. Machine 1 produces 60% of items and has a 3% defect rate; Machine 2 produces 40% of items and has a 5% defect rate. An item is selected at random.

(a) Compute the probability that the item is defective.
(b) Given the item is defective, compute the probability it came from Machine 1.
(c) A second item is drawn (independently) and is also defective. Update the posterior for Machine 1 using Bayes' theorem applied to both observations.

Exercise 6.4 (Independence Structure in MDPs). $(**)$
Consider a Markov decision process with state space $\mathcal{S} = \{s_1, s_2\}$, action space $\mathcal{A} = \{a_1, a_2\}$, and deterministic policy $\pi(s_1) = a_1$, $\pi(s_2) = a_2$. The transition probabilities are $\mathbb{P}(S_{t+1} = s_1 \mid S_t = s_1, A_t = a_1) = 0.7$ and $\mathbb{P}(S_{t+1} = s_1 \mid S_t = s_2, A_t = a_2) = 0.4$.

(a) Compute $\mathbb{P}(S_2 = s_1)$ given $\mathbb{P}(S_0 = s_1) = 1/2$ using the law of total probability (applied twice).
(b) Are $\{S_0 = s_1\}$ and $\{S_2 = s_1\}$ independent? Compute $\mathbb{P}(S_0 = s_1, S_2 = s_1)$ and compare with $\mathbb{P}(S_0 = s_1)\mathbb{P}(S_2 = s_1)$.

Exercise 6.5 (Constructing Mutual Independence from Pairwise). $(***)$
For $n = 4$, construct a finite probability space and events A_1, A_2, A_3, A_4 that are pairwise independent but not mutually independent. Verify all $\binom{4}{2} = 6$ pairwise conditions and exhibit at least one subset of size ≥ 3 for which the product formula fails.

Hint: Consider a sample space with $2^3 = 8$ equally likely outcomes. Define each A_i as a union of exactly 4 outcomes, chosen so that every pair of events intersects in exactly 2 outcomes.

Exercise 6.6 (Conditional Probability as a Probability Measure). $(***)$
Let $(\Omega, \mathbb{P})$ be a probability space and B an event with $\mathbb{P}(B) > 0$. Define $\mathbb{P}_B(A) = \mathbb{P}(A \mid B)$.

(a) Prove that $\mathbb{P}_B$ satisfies all three Kolmogorov axioms (this was stated in Remark 4.2; provide the full proof).
(b) Prove that if $A_1 \subseteq A_2 \subseteq \cdots$, then $\mathbb{P}_B(\bigcup_n A_n) = \lim_{n\to\infty} \mathbb{P}_B(A_n)$ (continuity of conditional probability from below).
(c) Give an example showing that the continuity from above can fail if we allow $\mathbb{P}(B) = 0$, motivating the need for measure-theoretic conditional expectation (Phase 01).

7 Summary

- A *probability space* $(\Omega, \mathbb{P})$ assigns probabilities to events via three axioms: non-negativity, normalization ($\mathbb{P}(\Omega) = 1$), and countable additivity.
- The *complement rule* ($\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$) and *monotonicity* ($A \subseteq B \Rightarrow \mathbb{P}(A) \leq \mathbb{P}(B)$) follow immediately from the axioms.
- *Inclusion-exclusion* computes $\mathbb{P}(\bigcup A_i)$ via an alternating sum of intersection probabilities; truncating it gives the union bound $\mathbb{P}(\bigcup A_i) \leq \sum \mathbb{P}(A_i)$.
- *Continuity of probability* extends the axioms to infinite limits: $\mathbb{P}(\bigcup A_n) = \lim \mathbb{P}(A_n)$ for increasing sequences and $\mathbb{P}(\bigcap A_n) = \lim \mathbb{P}(A_n)$ for decreasing sequences.

- *Conditional probability* $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$ formalizes how partial information updates uncertainty; it is itself a probability measure.
- *Bayes' theorem* inverts the direction of conditioning: $\mathbb{P}(B_j \mid A) \propto \mathbb{P}(A \mid B_j)\,\mathbb{P}(B_j)$, combining prior beliefs with observed evidence.
- Events A and B are *independent* if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$; *mutual independence* of n events requires the product formula for all $2^n - n - 1$ subsets of size ≥ 2.
- Pairwise independence does not imply mutual independence: the product rule can hold for every pair yet fail for larger subsets.

8 Further Reading

- **Lesson 10** introduces random variables, probability mass functions, and expectation, building directly on the probability space framework developed here.
- **Lesson 11** develops common distributions (binomial, geometric, Poisson) and their properties using the tools of conditional probability and independence.
- **Lesson 12** treats continuous random variables and probability density functions, extending the discrete theory.
- **Phase 01, Lesson 1** (σ-algebras) replaces the power-set approach with σ-algebras, necessary for uncountable sample spaces. Phase 01, Lesson 4 revisits probability spaces in the measure-theoretic framework.
- For a more combinatorial treatment of conditional probability, see Blitzstein and Hwang [2019, Chapter 2], which includes many worked examples from genetics, medicine, and game theory.
- The historical development of the Kolmogorov axioms (1933) and their unification of discrete and continuous probability is discussed in Grimmett and Stirzaker [2020, Chapter 1]. Earlier approaches by Laplace, Bernoulli, and von Mises lacked the generality of the measure-theoretic framework.

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